

◦ Aim:

- To test a single-phase energy meter by direct loading.

◦ Apparatus:

Equipment	Range	Quantity
1- Φ Energy meter	1200 rev/kWh	1
1- Φ Autotransformer/Variac	0-260V	1
Wattmeter	75/150/300V, 5/10A	1
Voltmeter (AC/MI)	0-300V	1
Ammeter (AC/MI)	0-5A	1
Resistive load	17.50KW, 230V	1
Connecting wires	—	Required

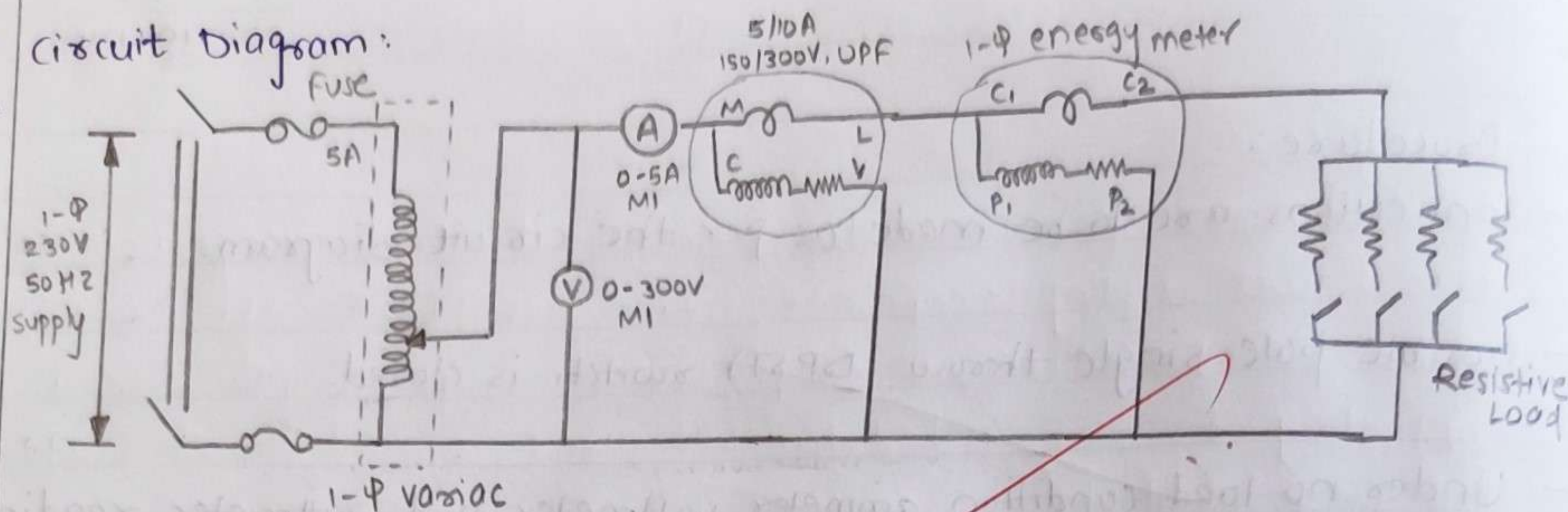
◦ Theory:

- Induction type of energy meters are universally used for measurement of energy in domestic and industrial AC circuits. Induction type of meters possesses lower friction and higher torque/weight ratio. Also, they are inexpensive and accurate, and retain their accuracy over a wide range of loads and temperature conditions.

- These are four main parts of the operating mechanism:

- (i) Driving system
- (ii) Moving system
- (iii) Braking system
- (iv) Registering system.

• Circuit Diagram:

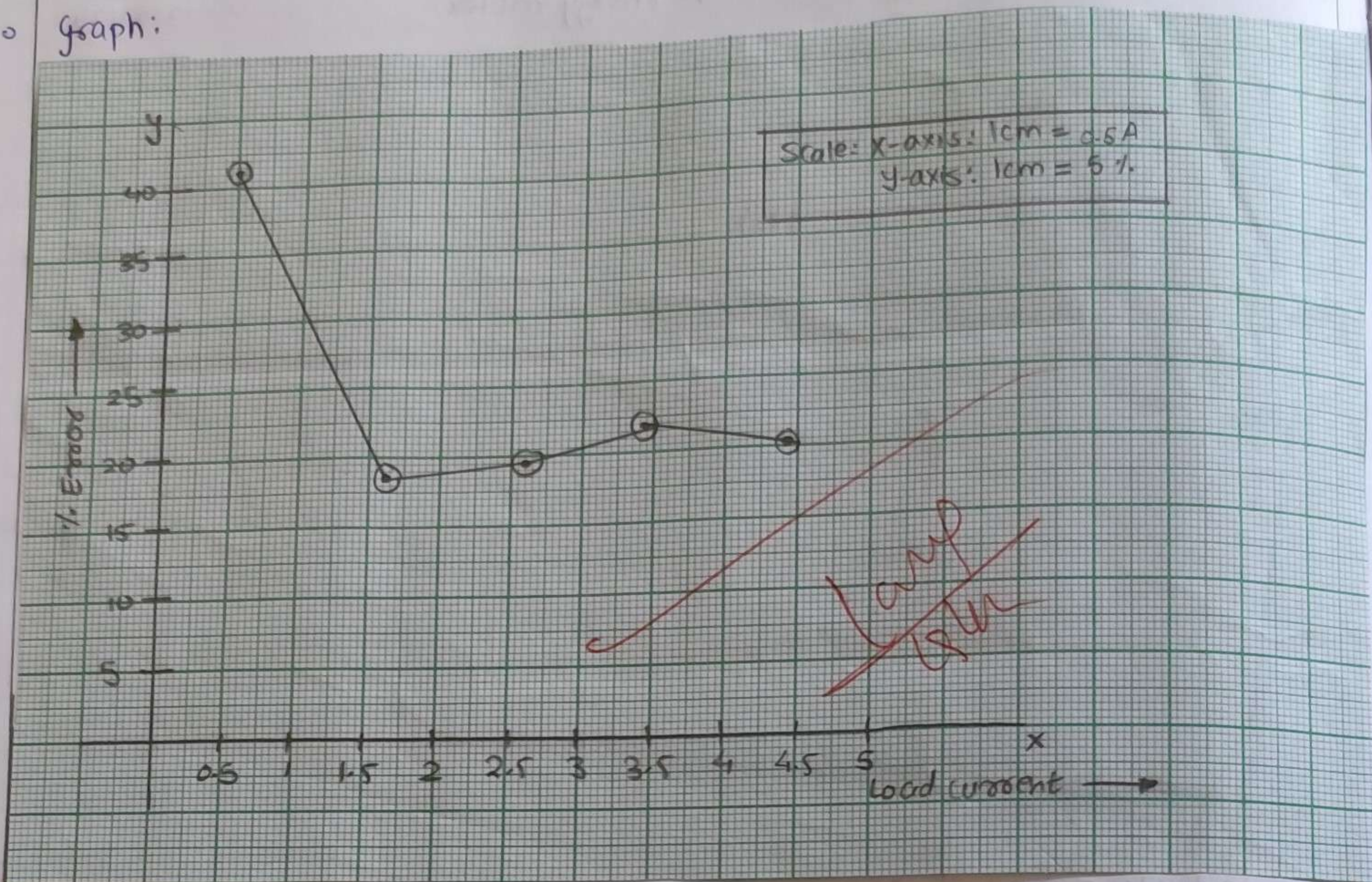


• observation Table:

Load (w)	Voltage V(V)	Current I(A)	Power P(W)	No. of revolutions (n)	Time t(s)	Indicated Energy $E_1 = \frac{n}{K} \times 10^3$ (Wh)	True Energy $E_2 = \frac{Pt}{3600}$ (Wh)	% Error $= \frac{E_2 - E_1}{E_1} \times 100$
250	230	0.5	135	20	265	16.67	9.9375	40.38
500	230	1.6	340	20	144	16.67	13.6	18.41
750	230	2.6	520	20	93	16.67	13.43	19.43
1000	230	3.4	650	20	72	16.67	13	22.01
1250	230	4.3	840	20	57	16.67	13.3	20.21

K is the energy meter constant = 1200 rev/kWh

Graph:



% Error vs load current graph.

- Theory (continued...):
- Driving system: The driving system of the meter consists of two electromagnets. The core of these electromagnets is made up of silicon steel laminations. The coil of one of the electromagnets is excited by the load current. This coil is called the 'current coil'. The coil of second electromagnet is connected across the supply and therefore carries a current proportional to the supply voltage. This coil is called as the 'pressure coil'. The function of copper shading bands at central limb is to bring the flux produced by the shunt magnet exactly in quadrature with the applied voltage.
- Moving system: This consists of an Aluminium disc mounted on a light alloy shaft. This disc is positioned in the air gap b/w series and shunt magnets.
- Braking system: A permanent magnet positioned near the edge of the aluminium disc forms the braking system. The Aluminium disc moves in the field of this magnet and thus provides a braking torque. The position of the magnet is adjustable, and hence braking torque can be adjusted by shifting the permanent magnet to different radial positions.
- Registering system: The function of registering system is to record a number continuously which is proportional to the revolutions to system.
- In all induction instruments, we have two fluxes produced by currents flowing in the windings of the instrument. These fluxes are alternating in nature and so they produce emfs in a metallic disc or a drum provided for the purpose. These emfs in turn circulate the eddy currents in

metallic disc or drum. The breaking torque is produced by the interaction of eddy current and field of permanent magnet. This torque is directly proportional to product of flux of magnet, magnitude of eddy current, and effective radius 'R' from axis of disc. The term testing includes the checking of the actual registration of meter as well as the adjustments done to bring the errors within prescribed limits.

Procedure:

- Connect the circuit as per the diagram.
- Keep the 1- Φ variac at zero-volt position.
- Now, switch on the power supply.
- Gradually vary the variac to apply the rated voltage (230V).
- For different values of load, note down the readings of the ammeter, voltmeter, wattmeter and time taken for 20 revolutions of the disc.
- Gradually vary the variac to zero volt position.
- Switch off the power supply.
- Calculate observed reading, actual reading, % error, % correction.
- Draw the graph between load current vs % Error.

Result:

- Single phase (1- Φ) energy meter is tested by direct loading.
- % Error vs load current graph is plotted.

*Work
done*